Week 2 Use-Case Descriptions

Traceability

These use cases are derived from:

- **US001:** View enrolment and thesis details
- **US004:** Update student status and thesis submission dates

They contribute to:

- **FR02:** Student Record Management
- **FR07:** Student Status Management
- **FR10:** Thesis Submission Management
- **BR12:** Student status values are restricted to defined institutional states

How to Use This Document

Each use case contains one supplied alternative flow. During the lab, identify additional alternative flows and update the descriptions before or alongside implementation.

The supplied domain model contains candidate fields. Where a use case refers to enrolment, thesis, status, or submission information, decide which domain fields support that behaviour. Do not assume that every supplied field must be implemented.

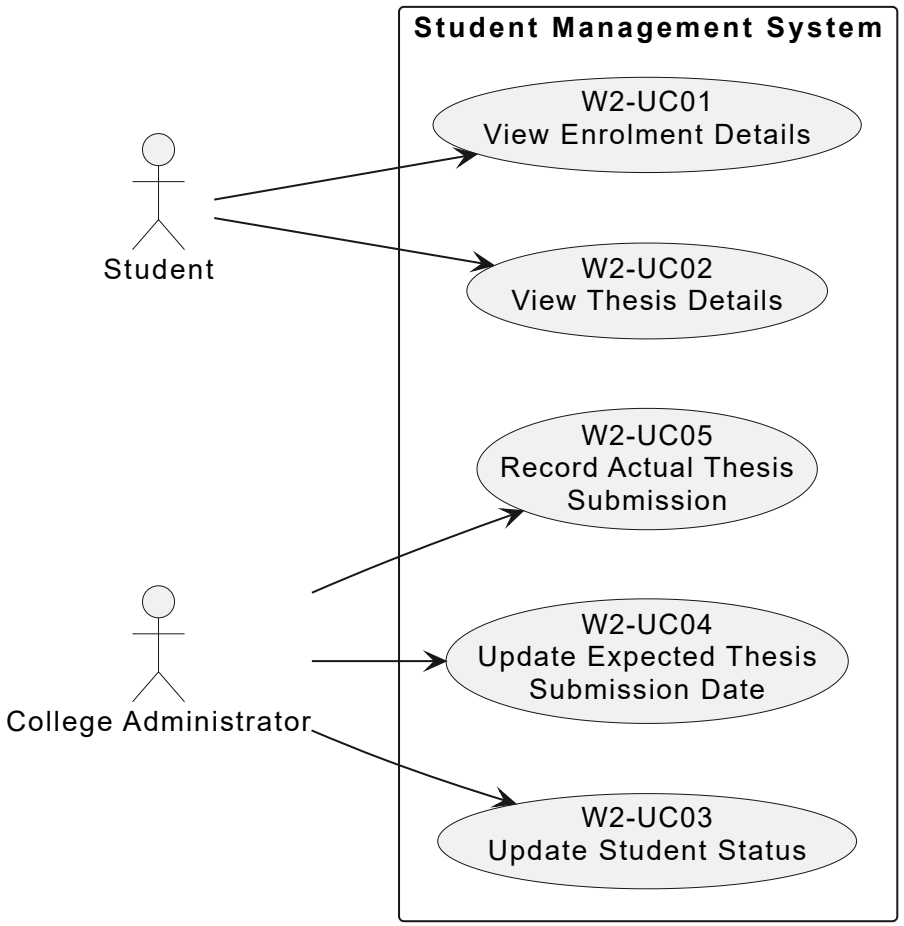
The Week 2 prototype uses an in-memory collection. Permanent persistence, authentication, role enforcement, and audit history are outside this lab's implementation scope.

Use-Case Summary

Use Case ID	Derived From	Use Case	Primary Actor	Goal	Supplied Alternative Flow
W2-UC01	US001	View Enrolment Details	Student	View current enrolment information, including programme, mode of study, start date, and status	Student Not Found
W2-UC02	US001	View Thesis Details	Student	View thesis information, including title and expected or actual submission dates	Thesis Information Not Available
W2-UC03	US004	Update Student Status	College Administrator	Change the current academic status of a student	Student Not Found

Use Case ID	Derived From	Use Case	Primary Actor	Goal	Supplied Alternative Flow
W2-UC04	US004	Update Expected Thesis Submission Date	College Administrator	Change the date by which a student is expected to submit their thesis	Student Not Found
W2-UC05	US004	Record Actual Thesis Submission	College Administrator	Record the actual thesis submission date and update the student status to Submitted	Invalid Submission Date

Week 2 - Student Management Use-Case Diagram



W2-UC01: View Enrolment Details

Derived From: US001
Primary Actor: Student

Goal

View the student's current enrolment information.

Preconditions

- The student can access the Student Management System.
- The student has a student number.

Trigger

The student selects **Enrolment Details** and supplies their student number.

Main Success Scenario

1. The student requests their enrolment details.
2. The system receives the student number.
3. The system searches the in-memory student collection.
4. The system finds the matching student record.
5. The system retrieves the available enrolment information.
6. The system displays:
    - Student number
    - Student name
    - Research programme
    - Mode of study
    - Start date
    - Current student status
7. The student reviews the information.

Alternative Flow A1: Student Not Found

At Step 4:

1. No student matches the supplied student number.
2. The system displays **Student not found**.
3. The use case ends.

Postconditions

Success

- The student's available enrolment details have been displayed.
- No student information has been changed.

Failure

- No student information has been changed.
- A student-not-found message has been displayed.

W2-UC02: View Thesis Details

Derived From: US001
Primary Actor: Student

Goal

View the student's thesis and submission information.

Preconditions

- The student can access the Student Management System.
- The student has a student number.

Trigger

The student selects **Thesis Details** and supplies their student number.

Main Success Scenario

1. The student requests their thesis details.
2. The system receives the student number.
3. The system searches the in-memory student collection.
4. The system finds the matching student record.
5. The system retrieves the available thesis information.
6. The system displays:
    - Thesis title
    - Expected submission date
    - Actual submission date, if recorded
    - Current student status
7. The student reviews the information.

Alternative Flow A1: Thesis Information Not Available

At Step 5:

1. The student record exists, but no thesis information has been recorded.
2. The system displays **Thesis information not available**.
3. The use case ends.

Postconditions

Success

- The student's available thesis details have been displayed.
- No student or thesis information has been changed.

Failure

- No student or thesis information has been changed.
- A thesis-information-not-available message has been displayed.

W2-UC03: Update Student Status

Derived From: US004
Primary Actor: College Administrator

Goal

Change the current academic status of a student.

Preconditions

- The College Administrator can access the Student Management System.
- The College Administrator can access the student status management function.

Trigger

The College Administrator selects **Edit Student Status**.

Main Success Scenario

1. The College Administrator supplies a student number.
2. The system searches the in-memory student collection.
3. The system finds the matching student.
4. The system displays the student's current status and the available status values.
5. The College Administrator selects a new status.
6. The College Administrator submits the change.
7. The system updates the student's `Status` property.
8. The system redirects to the student's details page.
9. The system displays the updated status.

Alternative Flow A1: Student Not Found

At Step 3:

1. No student matches the supplied student number.
2. The system displays **Student not found**.
3. The student's status is not changed.
4. The use case ends.

Postconditions

Success

- The student's status has been updated in the in-memory collection.
- The updated status remains available until the application stops or restarts.

Failure

- The student's status remains unchanged.
- A student-not-found message has been displayed.

Student Review Task

The supplied alternative flow is intentionally incomplete.

Identify at least two additional situations in which the update might not succeed. Document each situation as an additional alternative flow and state its postconditions.

Possible areas to consider include:

- A missing student number
- An invalid student number format
- A missing status
- An unsupported status value
- Selecting the student's existing status

W2-UC04: Update Expected Thesis Submission Date

Derived From: US004
Primary Actor: College Administrator

Goal

Update the date by which a student is expected to submit their thesis.

Preconditions

- The College Administrator can access the Student Management System.
- The College Administrator can access the thesis administration function.

Trigger

The College Administrator selects **Edit Thesis Information**.

Main Success Scenario

1. The College Administrator supplies a student number.
2. The system searches the in-memory student collection.
3. The system finds the matching student.
4. The system displays the existing expected thesis submission date.
5. The College Administrator enters a new expected submission date.
6. The College Administrator submits the change.
7. The system checks that the supplied value is a valid date.
8. The system updates the student's expected thesis submission date.
9. The system redirects to the student's thesis details page.
10. The system displays the updated date.

Alternative Flow A1: Student Not Found

At Step 3:

1. No student matches the supplied student number.
2. The system displays **Student not found**.
3. The expected submission date is not changed.
4. The use case ends.

Postconditions

Success

- The expected thesis submission date has been updated in the in-memory collection.
- The revised date remains available until the application stops or restarts.

Failure

- The previous expected submission date remains unchanged.
- A student-not-found message has been displayed.

W2-UC05: Record Actual Thesis Submission

Derived From: US004
Primary Actor: College Administrator

Goal

Record that a student has submitted their thesis and capture the actual submission date.

Preconditions

- The College Administrator can access the Student Management System.
- The College Administrator can access the thesis submission function.

Trigger

The College Administrator selects **Record Thesis Submission**.

Main Success Scenario

1. The College Administrator supplies a student number.
2. The system searches the static in-memory student collection.
3. The system finds the matching student.
4. The system displays the student's current thesis information.
5. The College Administrator enters the actual submission date.
6. The College Administrator submits the change.
7. The system checks that the supplied value is a valid date.
8. The system records the actual submission date on the `Student` object.
9. The system updates the student's status to `Submitted`.
10. The system redirects to the student's thesis details page.
11. The system displays the updated submission information.

Alternative Flow A1: Invalid Submission Date

At Step 7:

1. The supplied value is not a valid date.
2. The system does not update the actual submission date.
3. The system does not update the student's status.
4. The system redisplays the form with an **Invalid submission date** message.
5. The College Administrator may correct the value and resubmit the form.

Postconditions

Success

- The actual submission date has been recorded on the `Student` object.
- The student's status has been updated to `Submitted`.
- Both changes are available in the in-memory collection.
- The changes remain available until the application stops or restarts.

Failure

- The existing actual submission date remains unchanged.
- The student's existing status remains unchanged.
- An invalid-submission-date message has been displayed.

Student Review Task

The supplied alternative flow is intentionally incomplete.

Identify at least two additional situations in which the thesis submission might not be recorded successfully. Document each situation as an additional alternative flow and state its postconditions.

Possible areas to consider include:

- A missing student number
- An invalid student number format
- A student record that does not exist
- A missing submission date
- A submission date that is in the future
- A submission date that has already been recorded
- A submission date that conflicts with existing thesis information
- A student whose status cannot be changed to `Submitted`

Cross-Use-Case Design Decisions

Before implementing the use cases, decide:

- Which Student fields are required by each view and update operation.
- How expected and actual thesis submission dates are represented.
- Which status values are allowed.
- How optional information is displayed.
- Which additional validation rules are required.

When you add a validation rule, add or revise the corresponding alternative flow and postconditions.